EXTENSION 2

Stage 6

Calculus (Ext2), C1 Integration (Ext2)

Partial Fractions

Teacher: Nardene Dodd

Exam Equivalent Time: 100.5 minutes *(based on allocation of 1.5 minutes per mark)*

Questions

1. Calculus, EXT2 C1 2019 HSC 11d

   Find $\displaystyle\int \frac{6}{x^2-9}\,dx$. ***(3 marks)***

2. Calculus, EXT2 C1 2011 HSC 1c

   i. Find real numbers a, b and c such that

   $$\frac{1}{x^2(x-1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x-1}. \quad \textit{(2 marks)}$$

   ii. Hence, find $\displaystyle\int \frac{1}{x^2(x-1)}\,dx$ ***(2 marks)***

3. Calculus, EXT2 C1 2006 HSC 1c

   i. Given that $\dfrac{16x-43}{(x-3)^2(x+2)}$ can be written as

   $$\frac{16x-43}{(x-3)^2(x+2)} = \frac{a}{(x-3)^2} + \frac{b}{x-3} + \frac{c}{x+2},$$

   where a, b and c are real numbers, find a, b and c. ***(3 marks)***

   ii. Hence find $\displaystyle\int \frac{16x-43}{(x-3)^2(x+2)}\,dx$. ***(2 marks)***

4. Calculus, EXT2 C1 2005 HSC 1b

   i. Find real numbers a and b such that

   $$\frac{5x}{x^2-x-6} \equiv \frac{a}{x-3} + \frac{b}{x+2} \quad \textit{(2 marks)}$$

   ii. Hence find $\displaystyle\int \frac{5x}{x^2-x-6}\,dx$ ***(1 mark)***

5. Calculus, EXT2 C1 2022 HSC 12d

   Using partial fractions, evaluate $\displaystyle\int_2^n \frac{4+x}{(1-x)(4+x^2)}\,dx$, giving your answer in the form $\dfrac{1}{2}\ln\left(\dfrac{f(n)}{8(n-1)^2}\right)$, where $f(n)$ is a function of n. ***(4 marks)***

6. Calculus, EXT2 C1 2023 HSC 11f

   Find $\displaystyle\int_0^2 \frac{5x-3}{(x+1)(x-3)}\,dx$. ***(4 marks)***

7. Calculus, EXT2 C1 2018 HSC 11c

   By writing $\dfrac{x^2-x-6}{(x+1)(x^2-3)}$ in the form $\dfrac{a}{x+1} + \dfrac{bx+c}{x^2-3}$,

   find $\displaystyle\int \frac{x^2-x-6}{(x+1)(x^2-3)}\,dx$. ***(4 marks)***

8. Calculus, EXT2 C1 2021 HSC 11f

   Express $\dfrac{3x^2-5}{(x-2)(x^2+x+1)}$ as a sum of partial fractions over $\mathbb{R}$. ***(3 marks)***

9. Calculus, EXT2 C1 2007 HSC 1e

   It can be shown that

   $$\frac{2}{x^3+x^2+x+1} = \frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1}. \quad \text{(Do NOT prove this.)}$$

   Use this result to evaluate $\displaystyle\int_{\frac{1}{2}}^{2} \frac{2}{x^3+x^2+x+1}\,dx$. ***(4 marks)***

10. Calculus, EXT2 C1 2017 HSC 14a

It is given that $x^4+4=(x^2+2x+2)(x^2-2x+2)$.

i. Find A and B so that $\dfrac{16}{x^4+4}=\dfrac{A+2x}{x^2+2x+2}+\dfrac{B-2x}{x^2-2x+2}$. *(1 mark)*

ii. Hence, or otherwise, show that for any real number m

$$\int_0^m \frac{16}{x^4+4}\,dx=\ln\left(\frac{m^2+2m+2}{m^2-2m+2}\right)+2\tan^{-1}(m+1)+2\tan^{-1}(m-1).$$ *(2 marks)*

iii. Find the limiting value as $m\to\infty$ of

$$\int_0^m \frac{16}{x^4+4}\,dx.$$ *(1 mark)*

11. Calculus, EXT2 C1 2003 HSC 1d

i. Find the real numbers a and b such that

$$\frac{5x^2-3x+13}{(x-1)(x^2+4)}\equiv\frac{a}{x-1}+\frac{bx-1}{x^2+4}$$ *(2 marks)*

ii. Hence find $\displaystyle\int\frac{5x^2-3x+13}{(x-1)(x^2+4)}\,dx$ *(2 marks)*

12. Calculus, EXT2 C1 2008 HSC 1e

It can be shown that

$$\frac{8(1-x)}{(2-x^2)(2-2x+x^2)}=\frac{4-2x}{2-2x+x^2}-\frac{2x}{2-x^2}$$ (do NOT prove this)

Use this result to evaluate $\displaystyle\int_0^1\frac{8(1-x)}{(2-x^2)(2-2x+x^2)}\,dx$ *(4 marks)*

13. Calculus, EXT2 C1 2022 SPEC1 4

Find $\displaystyle\int\frac{3x^2+4x+12}{x(x^2+4)}\,dx$. *(4 marks)*

14. Calculus, EXT2 C1 EQ-Bank 1

Using partial fractions, show that

$$\int_0^{\frac{1}{2}}\frac{8}{1-x^4}\,dx=\log_e 9-4\tan^{-1}\left(\frac{1}{2}\right)$$ *(3 marks)*

15. Calculus, EXT2 C1 2009 HSC 1d

Evaluate $\displaystyle\int_2^5\frac{x-6}{x^2+3x-4}\,dx.$ *(4 marks)*

16. Calculus, EXT2 C1 2010 HSC 1c

Find $\displaystyle\int\frac{1}{x(x^2+1)}\,dx$. *(3 marks)*

17. Calculus, EXT2 C1 2010 HSC 5b

Show that $\displaystyle\int\frac{dy}{y(1-y)}=\ln\left(\frac{y}{1-y}\right)+c$

for some constant c, where $0<y<1$. *(2 marks)*

18. Calculus, EXT2 C1 2011 HSC 7b

Let $\displaystyle I=\int_1^3\frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)}\,dx.$

i. Use the substitution $u=4-x$ to show that

$$I=\int_1^3\frac{\sin^2\left(\frac{\pi}{8}u\right)}{u(4-u)}\,du.$$ *(2 marks)*

ii. Hence, find the value of I. *(3 marks)*

Worked Solutions

1. Calculus, EXT2 C1 2019 HSC 11d

Using partial fractions:

$$\begin{aligned}\frac{6}{x^2-9} &= \frac{6}{(x+3)(x-3)} \\ &= \frac{A}{x+3}+\frac{B}{x-3}\end{aligned}$$

$$\therefore A(x-3)+B(x+3)=6$$

When $x=3,\ 6B=6 \Rightarrow\ B=1$
When $x=-3,\ -6A=6 \Rightarrow\ A=-1$

$$\begin{aligned}\int \frac{6}{x^2-9}\,dx &= \int \frac{-1}{x+3}+\frac{1}{x-3}\,dx \\ &= -\ln|\,x+3\,|+\ln|\,x-3\,|+C \\ &= \ln\left|\,\frac{x-3}{x+3}\,\right|+C\end{aligned}$$

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2. Calculus, EXT2 C1 2011 HSC 1c

i. $\dfrac{1}{x^2(x-1)}=\dfrac{a}{x}+\dfrac{b}{x^2}+\dfrac{c}{x-1}$

$1=ax(x-1)+b(x-1)+cx^2$
$1=ax^2+cx^2-ax+bx-b$
$1=(a+c)x^2+(b-a)x-b$

Equating coefficients:
$\Rightarrow 0=a+c,\quad b-a=0,\quad -b=1$
$\therefore a=-1,\quad b=-1,\quad c=1$

ii. $$\begin{aligned}\int \frac{1}{x^2(x-1)}\,dx &= \int\left(-\frac{1}{x}-\frac{1}{x^2}+\frac{1}{x-1}\right)dx \\ &= -\log_e x+\frac{1}{x}+\log_e(x-1)+c \\ &= \log_e\left(\frac{x-1}{x}\right)+\frac{1}{x}+c\end{aligned}$$

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3. Calculus, EXT2 C1 2006 HSC 1c

i. $\dfrac{16x-43}{(x-3)^2(x+2)}=\dfrac{a}{(x-3)^2}+\dfrac{b}{x-3}+\dfrac{c}{x+2}$

$16x-43=a(x+2)+b(x-3)(x+2)+c(x-3)^2$

When $x=3,\quad 5a=5\quad \Rightarrow a=1$
When $x=-2,\quad 25c=-75\quad \Rightarrow c=-3$
When $x=0$

$-43=2(1)-6b+(-3)(-3)^2$
$6b=18$
$b=3$

$\therefore a=1, b=3, c=-3$

ii. $$\begin{aligned}&\int \frac{16x-43}{(x-3)^2(x+2)}\,dx \\ &= \int\left(\frac{1}{(x-3)^2}+\frac{3}{x-3}-\frac{3}{x+2}\right)dx \\ &= -\frac{1}{x-3}+3\ln(x-3)-3\ln(x+2)+c \\ &= -\frac{1}{x-3}+3\ln\left(\frac{x-3}{x+2}\right)+c\end{aligned}$$

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4. Calculus, EXT2 C1 2005 HSC 1b

i. $\dfrac{5x}{x^2-x-6} \equiv \dfrac{a(x+2)+b(x-3)}{(x-3)(x+2)}$

Equating numerators:

$5x = ax + 2a + bx - 3b$

$5x = (a+b)x + 2a - 3b$

$a + b = 5 \ \ldots (1)$

$2a - 3b = 0 \ldots (2)$

$(1) \times 2 - (2)$

$5b = 10$

$b = 2$

$a = 3$

ii. $\displaystyle\int \frac{5x}{x^2-x-6}\,dx = \int \frac{3}{x-3}\,dx + \int \frac{2}{x+2}\,dx$

$= 3\log_e|x-3| + 2\log_e|x+2| + C$

5. Calculus, EXT2 C1 2022 HSC 12d

$$\frac{4+x}{(1-x)(4+x^2)} \equiv \frac{A}{1-x} + \frac{Bx+C}{4+x^2}$$

$$4+x \equiv A(4+x^2) + (Bx+C)(1-x)$$

If $x=1,\ 5=5A \ \Rightarrow \ A=1$

$(4+x) \equiv 4 + x^2 + Bx - Bx^2 + C - Cx$

$4 + x \equiv (1-B)x^2 + (B-C)x + C + 4$

$\Rightarrow B = 1,\ C = 0$

$$\begin{aligned}
\therefore &\int_2^n \frac{4+x}{(1-x)(4+x^2)}dx \\
&= \int_2^n \frac{1}{1-x} + \frac{x}{4+x^2}\,dx \\
&= \left[-\ln|1-x| + \frac{1}{2}\ln(4+x^2)\right]_2^n \\
&= -\ln|1-n| + \frac{1}{2}\ln(4+n^2) + \ln|1-2| - \frac{1}{2}\ln(4+2^2) \\
&= -\frac{1}{2}\ln(1-n)^2 + \frac{1}{2}\ln(4+n^2) - \frac{1}{2}\ln(8) \\
&= \frac{1}{2}\ln\left(\frac{4+n^2}{8(1-n^2)}\right)
\end{aligned}$$

Mean mark 85%.

6. Calculus, EXT2 C1 2023 HSC 11f

Let $\dfrac{5x-3}{(x+1)(x-3)}=\dfrac{A}{x+1}+\dfrac{B}{x-3}$ for $A, B \in \mathbb{R}$

$$\begin{aligned}\frac{5x-3}{(x+1)(x-3)} &= \frac{A}{x+1}+\frac{B}{x-3}\\ &= \frac{A(x-3)+B(x+1)}{(x+1)(x-3)}\end{aligned}$$

Equating numerators:

$A(x-3)+B(x+1)=5x-3$

When $x=3, 4B=12 \Rightarrow B=3$

When $x=-1, -4A=-8 \Rightarrow A=2$

$$\begin{aligned}\int_0^2 \frac{5x-3}{(x+1)(x-3)}\,dx &= \int_0^2 \frac{2}{x+1}+\frac{3}{x-3}\,dx\\ &= \Big[2\ln|x+1|+3\ln|x-3|\Big]_0^2\\ &= 2\ln 3+3\ln 1-2\ln 1-3\ln 3\\ &= -\ln 3\end{aligned}$$

7. Calculus, EXT2 C1 2018 HSC 11c

$$\frac{x^2-x-6}{(x+1)(x^2-3)} \equiv \frac{a}{x+1}+\frac{bx+c}{x^2-3}$$

$$x^2-x-6 \equiv a(x^2-3)+(bx+c)(x+1)$$

When $x=-1$

$1+1-6=-2a \quad \Rightarrow \quad a=2$

Equating co-efficients of x^2

$1=(a+b)x^2 \quad \Rightarrow \quad b=-1$

Equating co-efficients of x

$-1=b+c \quad \Rightarrow \quad c=0$

$$\begin{aligned}\therefore \int \frac{x^2-x-6}{(x+1)(x^2-3)}\,dx &= \int \frac{2}{x+1}-\frac{x}{x^2-3}\,dx\\ &= 2\ln|\,x+1\,|-\frac{1}{2}\ln|\,x^2-3\,|+c\end{aligned}$$

8. Calculus, EXT2 C1 2021 HSC 11f

$$\frac{3x^2-5}{(x-2)(x^2+x+1)}=\frac{A}{(x-2)}+\frac{Bx+C}{(x^2+x+1)}$$

$A(x^2+x+1)+(Bx+C)(x-2) \equiv 3x^2-5$

If $x=2$,

$7A=7 \Rightarrow A=1$

If $x=0$,

$1-2C=-5 \Rightarrow C=3$

Equating coefficients:

$x^2+x+1+Bx^2-2Bx+3x-6 \equiv 3x^2-5$

$(B+1)x^2+(4-2B)x-5 \equiv 3x^2-5$

$\Rightarrow B=2$

$$\therefore \frac{3x^2-5}{(x-2)(x^2+x+1)}=\frac{1}{x-2}+\frac{2x+3}{x^2+x+1}$$

9. Calculus, EXT2 C1 2007 HSC 1e

$$\begin{aligned}&\int_{\frac{1}{2}}^2 \frac{2}{x^3+x^2+x+1}\,dx\\ &= \int_{\frac{1}{2}}^2 \left(\frac{1}{x+1}-\frac{x}{x^2+1}+\frac{1}{x^2+1}\right)dx\\ &= \left[\log_e(x+1)-\frac{1}{2}\log_e(x^2+1)+\tan^{-1}x\right]_{\frac{1}{2}}^2\\ &= \left[\left(\log_e 3-\frac{1}{2}\log_e 5+\tan^{-1}2\right)-\left(\log_e\frac{3}{2}-\frac{1}{2}\log_e\frac{5}{4}+\tan^{-1}\frac{1}{2}\right)\right]\\ &= \log_e\frac{3}{\sqrt{5}}-\log_e\left(\frac{3}{2}\times\frac{2}{\sqrt{5}}\right)+\tan^{-1}2-\tan^{-1}\frac{1}{2}\\ &= \log_e\left(\frac{3}{\sqrt{5}}\times\frac{\sqrt{5}}{3}\right)+\tan^{-1}2-\tan^{-1}\frac{1}{2}\\ &= \tan^{-1}2-\tan^{-1}\frac{1}{2}\end{aligned}$$

10. Calculus, EXT2 C1 2017 HSC 14a

i. $\dfrac{16}{x^4+4} = \dfrac{A+2x}{x^2+2x+2} + \dfrac{B-2x}{x^2-2x+2}$

Multiply each side by x^4+4

$16 = (A+2x)(x^2-2x+2) + (B-2x)(x^2+2x+2)$

When $x=0$,

$2A+2B=16$

$A+B=8\ \ldots(1)$

When $x=1$

$(A+2)(1)+(B-2)(5)=16$

$A+2+5B-10=16$

$A+5B=24\ \ldots(2)$

Subtract $(2)-(1)$

$4B=16 \ \Rightarrow\ B=4$

$A=4$

ii. $\displaystyle\int_0^m \frac{16}{x^4+4}$

$$= \int_0^m \frac{4+2x}{x^2+2x+2}dx + \int_0^m \frac{4-2x}{x^2-2x+2}dx$$

$$= \int_0^m \frac{2x+2}{x^2+2x+2} + \frac{2}{x^2+2x+2}dx + \int_0^m \frac{2}{x^2-2x+2} - \frac{2x-2}{x^2-2x+2}dx$$

$$= \left[\ln(x^2+2x+2)\right]_0^m + \int_0^m \frac{2}{1+(x+1)^2}dx + \int_0^m \frac{2}{1+(x-1)^2}dx - \left[\ln(x^2-2x+2)\right]_0^m$$

$$= \left[\ln(m^2+2m+2) - \ln 2\right] + \left[2\tan^{-1}(x+1)\right]_0^m + \left[2\tan^{-1}(x-1)\right]_0^m - \left[\ln(m^2-2m+2) - \text{l}\right.$$

$$= \ln\left(\frac{m^2+2m+2}{m^2-2m+2}\right) + 2\left[\tan^{-1}(m+1) - \tan^{-1}(1)\right] + 2\left[\tan^{-1}(m-1) - \tan^{-1}(1)\right]$$

$$= \ln\left(\frac{m^2+2m+2}{m^2-2m+2}\right) + 2\tan^{-1}(m+1) - 2\cdot\frac{\pi}{4} + 2\tan^{-1}(m-1) + 2\cdot\frac{\pi}{4}$$

$$= \ln\left(\frac{m^2+2m+2}{m^2-2m+2}\right) + 2\tan^{-1}(m+1) + 2\tan^{-1}(m-1)$$

iii. $I = \displaystyle\int_0^m \frac{16}{x^4+4} = \ln\left(\frac{1+\frac{2}{m}+\frac{2}{m^2}}{1-\frac{2}{m}+\frac{2}{m^2}}\right) + 2\tan^{-1}(m+1) + 2\tan^{-1}(m-1)$

$$\lim_{m\to\infty} I = \ln 1 + 2\cdot\frac{\pi}{2} + 2\cdot\frac{\pi}{2}$$

$$= 0+\pi+\pi$$

$$= 2\pi$$

11. Calculus, EXT2 C1 2003 HSC 1d

i. $\dfrac{5x^2-3x+13}{(x-1)(x^2+4)} \equiv \dfrac{a(x^2+4)+(bx-1)(x-1)}{(x-1)(x^2+4)}$

Equating numerators:

$5x^2-3x+13 = ax^2+4a+bx^2-bx-x+1$

$= (a+b)x^2 + (-b-1)x + 4a + 1$

$-b-1=-3 \ \Rightarrow\ b=2$

$a=3$

ii. $\displaystyle\int \frac{5x^2-3x+13}{(x-1)(x^2+4)}\,dx = \int \frac{3}{x-1}\,dx - \int \frac{2x}{x^2+4}\,dx - \int \frac{1}{4+x^2}\,dx$

$$= 3\log_e|x-1| - \log_e|x^2+4| - \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + C$$

12. Calculus, EXT2 C1 2008 HSC 1e

$$\int_0^1 \frac{8(1-x)}{(2-x^2)(2-2x+x^2)}\,dx$$

$$= \int_0^1 \frac{4-2x}{2-2x+x^2}\,dx - \int_0^1 \frac{2x}{2-x^2}\,dx$$

$$= \int_0^1 \frac{2-(2x-2)}{2-2x+x^2}\,dx + \left[\log_e|2-x^2|\right]_0^1$$

$$= \int_0^1 \frac{2}{1+(1-x)^2} - \int_0^1 \frac{2x-2}{2-2x+x^2} + \left[\log_e|2-x^2|\right]_0^1$$

$$= \left[-2\tan^{-1}(1-x)\right]_0^1 - \left[\log_e(2-2x+x^2)\right]_0^1 + \left[\log_e|2-x^2|\right]_0^1$$

$$= -2\tan^{-1}0 + 2\tan^{-1}1 - (\log_e 1 - \log_e 2) + (\log_e 1 - \log_e 2)$$

$$= 0 + 2\times\frac{\pi}{4} + \log_e 2 - \log_e 2$$

$$= \frac{\pi}{2}$$

13. Calculus, EXT2 C1 2022 SPEC1 4

Mean mark 55%.

Using partial fractions:

$$\frac{3x^2+4x+12}{x(x^2+4)} \equiv \frac{A}{x}+\frac{Bx+C}{x^2+4}$$

$$3x^2+4x+12 = A\left(x^2+4\right)+x(Bx+C)$$

$$3x^2+4x+12 = (A+B)x^2+Cx+4A$$

$4A = 12 \quad \Rightarrow \quad A = 3$

$C = 4$

$A+B = 3 \quad \Rightarrow \quad B = 0$

$$\therefore \int \frac{3x^2+4x+12}{x(x^2+4)}\,dx = \int \frac{3}{x}+\frac{4}{4+x^2}\,dx$$

$$= 3\ln|x|+2\tan^{-1}\left(\frac{x}{2}\right)+c$$

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14. Calculus, EXT2 C1 EQ-Bank 1

Using partial fractions:

$$\frac{8}{1-x^4} = \frac{A}{1-x^2}+\frac{B}{1+x^2}$$

$$A\left(1+x^2\right)+B\left(1-x^2\right) = 8$$

$$A+B+(A-B)x^2 = 8$$

$A+B = 8 \ \ldots (1)$

$A-B = 0 \ \ldots (2)$

$A = 4, \ B = 4$

$$\frac{4}{1-x^2} = \frac{A}{1-x}+\frac{B}{1+x}$$

$$A(1+x)+B(1-x) = 4$$

$$A+B+(A-B)x = 4$$

$A+B = 4 \ \ldots (1)$

$A-B = 0 \ \ldots (2)$

$A = 2, B = 2$

$$\int_0^{\frac{1}{2}} \frac{8}{1-x^4}\,dx = \int_0^{\frac{1}{2}} \frac{2}{1-x}+\frac{2}{1+x}+\frac{4}{1+x^2}\,dx$$

$$= \left[-2\ln|1-x|+2\ln|1+x|+4\tan^{-1}x\right]_0^{\frac{1}{2}}$$

$$= \left[2\ln\left|\frac{1+x}{1-x}\right|+4\tan^{-1}x\right]_0^{\frac{1}{2}}$$

$$= 2\ln\left|\frac{1\frac{1}{2}}{\frac{1}{2}}\right|+4\tan^{-1}\left(\frac{1}{2}\right)-\left(2\ln 1+4\tan^{-1}0\right)$$

$$= 2\ln 3+4\tan^{-1}\left(\frac{1}{2}\right)$$

$$= \ln 9+4\tan^{-1}\left(\frac{1}{2}\right)$$

15. Calculus, EXT2 C1 2009 HSC 1d

Using partial fractions:

$$\frac{x-6}{x^2+3x-4} = \frac{x-6}{(x+4)(x-1)}$$
$$= \frac{A}{x+4} + \frac{B}{x-1}$$
$$\therefore x-6 = A(x-1) + B(x+4)$$

When $x = 1,\ \ 5B = -5 \Rightarrow B = -1$
When $x = -4,\ \ -5A = -10 \Rightarrow A = 2$

$$\therefore \int_2^5 \frac{x-6}{x^2+3x-4}\,dx = \int_2^5 \frac{2}{x+4}\,dx - \int_2^5 \frac{dx}{x-1}$$
$$= [2\ln(x+4)]_2^5 - [\ln(x-1)]_2^5$$
$$= 2(\ln 9 - \ln 6) - (\ln 4 - \ln 1)$$
$$= 2\ln\left(\frac{3}{2}\right) - \ln 4$$
$$= 2\ln\left(\frac{3}{2}\right) - 2\ln 2$$
$$= 2\ln\left(\frac{3}{4}\right)$$

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16. Calculus, EXT2 C1 2010 HSC 1c

Using partial fractions:

$$\frac{1}{x(x^2+1)} = \frac{a}{x} + \frac{bx+c}{x^2+1}$$
$$1 = a\left(x^2+1\right) + x(bx+c)$$
$$1 = (a+b)x^2 + cx + a$$
$$\therefore c = 0,\ \ a = 1,\ \ b = -1$$

$$\therefore \int \frac{1}{x(x^2+1)}\,dx = \int \left(\frac{1}{x} - \frac{x}{x^2+1}\right)\,dx$$
$$= \ln|x| - \frac{1}{2}\ln\left(x^2+1\right) + c$$

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17. Calculus, EXT2 C1 2010 HSC 5b

Solution 1

$$\frac{d}{dy}\left[\ln\left(\frac{y}{1-y}\right) + c\right] = \frac{d}{dy}[\ln y - \ln(1-y) + c]$$
$$= \frac{1}{y} - \frac{-1}{1-y} + 0$$
$$= \frac{1-y+y}{y(1-y)}$$
$$= \frac{1}{y(1-y)}$$

Solution 2
Using partial fractions:

$$\frac{1}{y(1-y)} = \frac{A}{y} + \frac{B}{1-y}$$

$\Rightarrow A(y-1) + By = 1$
When $y = 1,\ B = 1$
When $y = 0,\ A = 1$

$$\therefore \int \frac{dy}{y(1-y)} = \int \left(\frac{1}{y} + \frac{1}{1-y}\right)\,dy$$
$$= \ln y - \ln(1-y) + c$$
$$= \ln\left(\frac{y}{1-y}\right) + c, \qquad \frac{y}{1-y} > 0 \ \text{ for } \ 0 < y < 1.$$

18. Calculus, EXT2 C1 2011 HSC 7b

i. Let $u = 4 - x, \quad du = -\,dx, \quad x = 4 - u$

When $x = 1, \quad u = 3$

When $x = 3, \quad u = 1$

$$\begin{aligned} I &= \int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)}\ dx \\ &= \int_3^1 \frac{\cos^2\left(\frac{\pi}{8}(4-u)\right)}{(4-u)u}(-\,du) \\ &= -\int_3^1 \frac{\cos^2\left(\frac{\pi}{2} - \frac{\pi}{8}u\right)}{(4-u)u}\ du \\ &= \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}u\right)}{u(4-u)}\ du \end{aligned}$$

ii. Since $\displaystyle\int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)}\ dx = \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}x\right)}{x(4-x)}\ dx$

We can add the integrals such that

$$\begin{aligned} 2I &= \int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)}dx + \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}x\right)}{x(4-x)}dx \\ &= \int_1^3 \frac{1}{x(4-x)}dx \end{aligned}$$

Using partial fractions:

♦ *Mean mark 36%.*

$$\begin{aligned} \frac{1}{x(4-x)} &= \frac{A}{x} + \frac{B}{4-x} \\ 1 &= A(4-x) + Bx \end{aligned}$$

When $x = 0, \quad A = \dfrac{1}{4}$

When $x = 0, \quad B = \dfrac{1}{4}$

$$\begin{aligned} 2I &= \frac{1}{4}\int_1^3 \left(\frac{1}{x} + \frac{1}{4-x}\right)\ dx \\ &= \frac{1}{4}\left[\log_e x - \log_e(4-x)\right]_1^3 \\ &= \frac{1}{4}\left[\log_e 3 - \log_e 1 - (\log_e 1 - \log_e 3)\right] \\ &= \frac{1}{2}\log_e 3 \end{aligned}$$

$$\therefore I = \frac{1}{4}\log_e 3$$